

f00-sh

joule distributed cluster

STANDARD OPERATING PROCEDURE - joule cluster

Document: JOULE-SOP-OPS-001

Date: 2026-08-04

Version: v0.1.13+

Standard: NASA NAI 1450.3 memorandum layout

Reply to Attn of: OPS/JOULE

TO: joule operators and release owners

FROM: Project Engineering

SUBJECT: joule operator standard operating procedure

REF: AGENTS.md, README.md, docs/design/cluster-v0.md,
docs/adr/0003-cuda-production-engine.md

Operators run joule as a GUI-first local pool or multi-agent internet cluster: dumb-user start path, production CUDA engine with content-proof weights, band-only peer-seeded shards (not full multi-TB on every box), fleet honesty for full Kimi, contribute-to-consume economy, and ship only when cargo gates and the docs pack are green.

1. Purpose

This SOP defines day-to-day operation of a joule lab or standing pool: quality gates, dumb-user join path, donor contribution control, full-Kimi fleet honesty, live capacity visibility (CLI and GUI), production smoke, and the release ship flow for f00-sh/joule.

2a. Native installers

Prefer OS installers from permanent site links or GitHub Releases. Files live on GitHub; joule.f00.sh only redirects (website only - not a weight CDN).

- Permanent: <https://joule.f00.sh/current/> and download.html
- Windows: joule-windows-x86_64-setup.exe (or versioned *-setup.exe)
- macOS: joule-darwin-aarch64.pkg or .dmg
- Linux: sudo dpkg -i joule-linux-x86_64.deb
- CLI: curl -fsSL <https://joule.f00.sh/current/install.sh> | sh

2b. Dumb-user local pool (preferred)

Do not invent ports for first light. Use the shipped one-shot path so control and agent come up together.

2b.1 Print checklist anytime.

```
joule get-started
```

2b.2 GUI path: run joule, click star DO EVERYTHING (local pool). First open auto-starts if :7700 free (JOULE_GUI_NO_AUTO=1 skips).

```
joule
```

2b.3 Headless one-shot control + agent.

```
joule start
```

2b.4 Reboot-safe user-session autostart (control + agent + tray).

```
joule service install
```

2b.5 Show Base URL + joule_ key + model for OpenAI clients.

```
joule connect
```

2b.6 Contributor chat (requires healthy agent for that account).

```
joule chat --prompt hi
```

2. Day-to-day operator gates

Before exposing any gateway or telling donors the pool is open, complete these checks in order. Fail closed on any red step.

2.1 Run workspace tests.

```
cargo test --workspace
```

2.2 Clippy with warnings denied.

```
cargo clippy --workspace --all-targets -- -D warnings
```

2.3 Release-binary economy smoke (2 agents, freeloader deny, contributor chat).

```
bash scripts/production-smoke.sh
```

2.4 Start local pool (GUI or start) and confirm capacity backends ≥ 1 .

```
joule start && joule capacity --api http://127.0.0.1:7700 --json
```

2.5 Attach a second agent for multi-node join when claiming a multi-backend pool.

2.6 Confirm GUI Overview shows backends and connection state; Graphs zoom/pan work.

Figure 1. Day-to-day operator gates

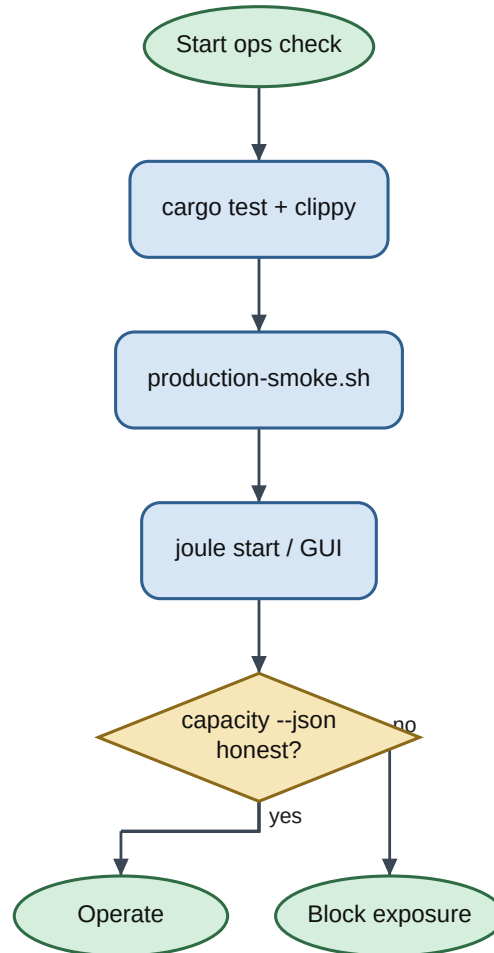


Figure 2. GUI-first operator surface



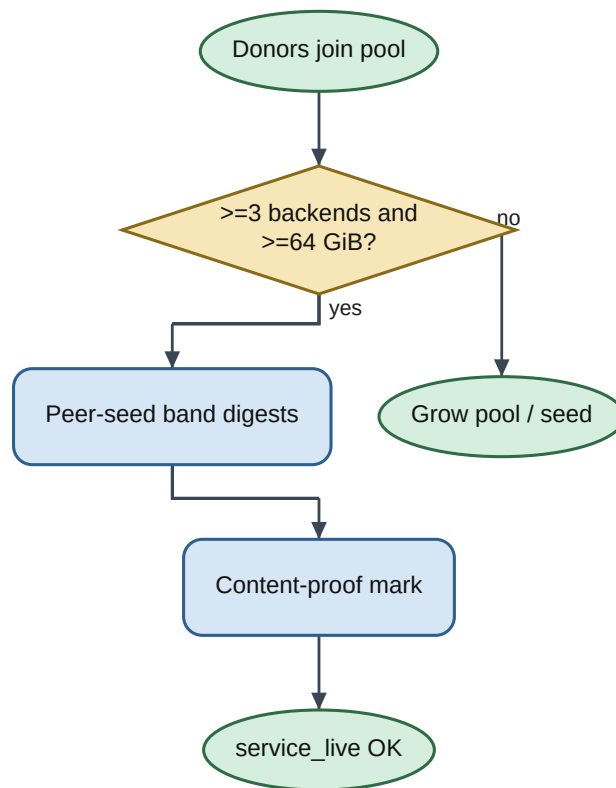
joule version and GUI tab path (Overview / Graphs / Donor / Chat) after release build.

3. Full Kimi honesty (do not overclaim)

Full Kimi is one logical device across donors. Never tell users every box must download multi-TB weights.

- Production quant: kimi-k3-shards - 96 real moonshotai/Kimi-K3 LFS digests in models/MANIFEST.json
- Band/shard residency only per donor; peer-seed by sha256; f00 is website only
- Fleet gate full_k3_service_fleet_ok: ≥ 64 GiB healthy aggregate VRAM and ≥ 3 backends
- Service digests SoT is production K3 (production_digests_ok); lab complete alone is not service_live
- Lab fixtures (lab-tiny/mid/large) = protocol CI only - not production Kimi quality
- Commit-gated quant upgrade: try_commit_* - fleet K3 recommend without bytes must not brick lab serve
- Agents use ProductionEngine (ADR 0003 CUDA driver FFI) with content-proof stage/infer

Figure 3. Full Kimi service-live path



4. Donor local controls (law 5)

Remote operators cannot raise local caps. Agents reload the policy path passed with --policy. Schedule windows use OS local timezone (tm_gmtoff on Unix).

- Pause/resume: joule donor pause|resume or GUI Donor tab / tray
- VRAM cap: joule donor set-cap N (0 clears) or tray --donor-set-cap
- Schedule / thermal / battery: donor policy JSON or GUI Donor tab
- Status: joule donor status - prints local schedule and sensors

5. Economy closed loop

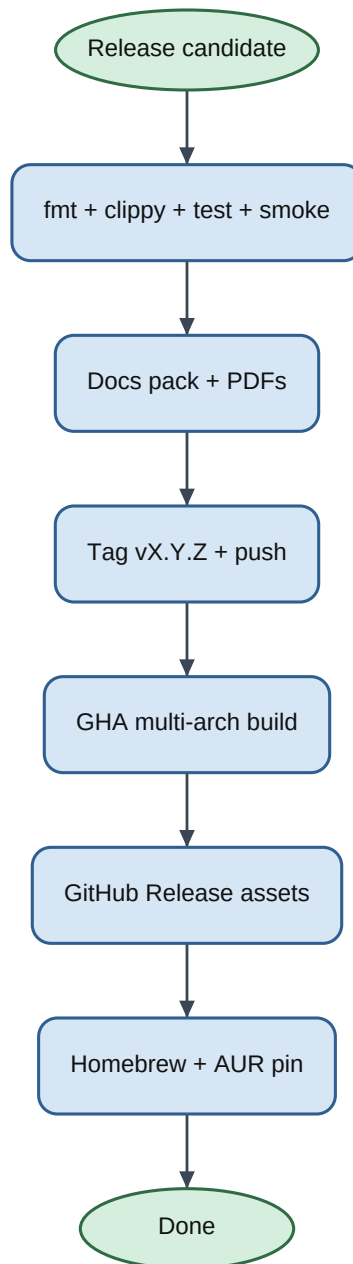
Public pool: no cash. Freeloaders without a healthy agent for the account get HTTP 401/403. Contributors chat on the OpenAI-shaped path after mint from contribution.

- Verify freeloader deny and contributor chat via production-smoke.sh
- Millijoules: eco=v0 fairness (?VRAM, tenure, leecher)
- Single model: kimi-open only

6. Release ship flow

Ship only when gates are green. Documentation pack is mandatory: CHANGELOG, file_id.diz, README, Pages, man, SOP PDF, this-version NASA memo PDF, SemVer tag, GitHub Release assets (CI on tag v*).

Figure 4. Release ship flow



7. Product laws (do not violate)

- Compute is the only currency on the public pool - no money
- No contribution ? no API access
- Credits = sealed hash-chain millijoules only; verified VRAM only for mint/placement
- Single model: kimi-open
- One logical device: N donors = one virtual GPU (sum of healthy VRAM)
- Website is not a weight CDN - content-addressed peer seed only
- Not every box downloads full multi-TB weights - band/shard slices

Project Engineering
f00-sh / joule OPS

Enclosure: file_id.diz; CHANGELOG.md; docs/releases/; ADR 0003
cc: Maintainers, Release Operators